AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-08-19 | Entry window 19 August 2026 | 22 records in scope, 13 rejected

22 RECORDS IN SCOPE (13 REJECTED OFF-TOPIC)	**9** SUBTOPIC CLUSTERS REPRESENTED	**6** ABSTRACT-FREE PUBLISHER DEPOSITS	**5** EFFECT ESTIMATES EXTRACTED	**3** REVIEWS / EVIDENCE SYNTHESES

Signal of the day

Cumulative exposure *duration*, not the day's concentration, is where today's hazard signal sits — and the day's largest acute study shows why.

Sivakumar & Kurian (*Environ Toxicol*) exposed female Wistar rats to 250 $\mu g/m^3$ $PM_{2.5}$, 3 h/day, for 1, 7, 14 or 21 days at a fixed concentration. At 1–7 days the cardiac mitochondrial response was **adaptive and fully reversible**: transient oxidative stress and quality-control activation, with respiration, ETC complex activity, ATP synthesis and ex vivo cardiac function all indistinguishable from control after a 24 h pollutant-free washout. From **14 days** the same daily dose produced coordinated suppression of respiration and ATP synthesis, mtDNA depletion and intramitochondrial metal accumulation — and these deficits **did not recover within the 24 h washout window**. Concentration was constant throughout; only duration changed. Set against this, Park et al. (*Gut Liver*) ran the day's largest acute design — 29,167 Korean cirrhosis patients, time-stratified case-crossover, 1 km daily $PM_{2.5}$ — and recovered **aOR 1.02 (95% CI 1.00–1.04) per IQR**, a lower bound sitting exactly on the null, with associations *attenuating* as the averaging window lengthened. Two designs, opposite time-scales, same message for instrumentation: a day-resolved mass metric is near its resolution floor, while the cumulative-duration metric that the animal work implicates is precisely what a continuously-logging personal or distributed sensor can deliver and a regulatory site cannot.

What changed today

- **Filtration works; the model that predicts it does not transfer.** Xia et al. measured 33 spaces in 12 child-serving facilities before and after enhanced HVAC filtration and portable air cleaners. $PM_{2.5}$ fell 45–54% in standard schools and the nursery (mixed-effects estimate ∼51% for the PACs alone), but barely moved in childcare houses. Crucially, single-zone mass-balance prediction held for regular classrooms and failed for portable buildings, very large rooms and houses repurposed for childcare. The buildings where the physics is unreliable are exactly the ones that need instrumenting.
- **Proteomics enters as an effect-modification layer, not a main-effect one.** He et al. screened 2,254 plasma proteins against LUR-modelled $PM_{2.5}$, PM_{coarse}, NO_2 and NO in 51,428 participants and report protein × pollutant *interactions* on epilepsy risk (CIT–$PM_{2.5}$, PTPRR–NO_2, BCL2L15–API), with hippocampal-volume and white-matter-hyperintensity correlates. The multiplicity burden here is roughly 9,000 interaction tests and the abstract reports no corrected significance level or interaction magnitude — treat as a candidate list, not an established mechanism.
- **$PM_{2.5}$ mass is again shown to be a poor proxy for a co-located particle property.** Yang et al. (Taiyuan, month-long winter campaign) found immersion-mode INP concentration rose from 0.61 to 7.47 L^{-1} (95% CI 6.64–8.41) during a desert-dust intrusion, yet showed $|r| < 0.3$ against sulfate, nitrate and OC and no significant covariation with any of five PMF-resolved $PM_{2.5}$ source factors that between them dominated the mass.
- **A clean emission-factor dataset for prescribed fire.** Blanco-Alegre et al. report $PM_{2.5}$ emission factors of 53.5 $g\,kg^{-1}$ (*Calluna*, flaming, MCE 90.6%) and 44.7 $g\,kg^{-1}$ (*Genista*, smouldering, MCE 70.8%), with OC and EC at 28.1% and 32.9% of $PM_{2.5}$ mass. Combustion-regime-resolved EFs of this quality are scarce for Mediterranean shrubland and directly usable in inventory work.
- **Negative result worth recording.** In Lahore's 2024–2025 smog season (mean $PM_{2.5}$ 96 $\mu g/m^3$, WHO 24 h guideline exceeded on every monitored day), routine mask use was *not* associated with lower symptom count — IRR 0.96 (0.84–1.09) — while current smoking was, at IRR 1.15 (1.00–1.32). Self-reported mask use in a cross-sectional sample cannot separate protection from reverse causation, but the null belongs on record.
- **Six of twenty-two records arrived without an abstract** — four Elsevier and two RSC deposits, carried with citation and metadata only, flagged as such throughout, and with no effect size inferred for any of them.

Corpus analytics

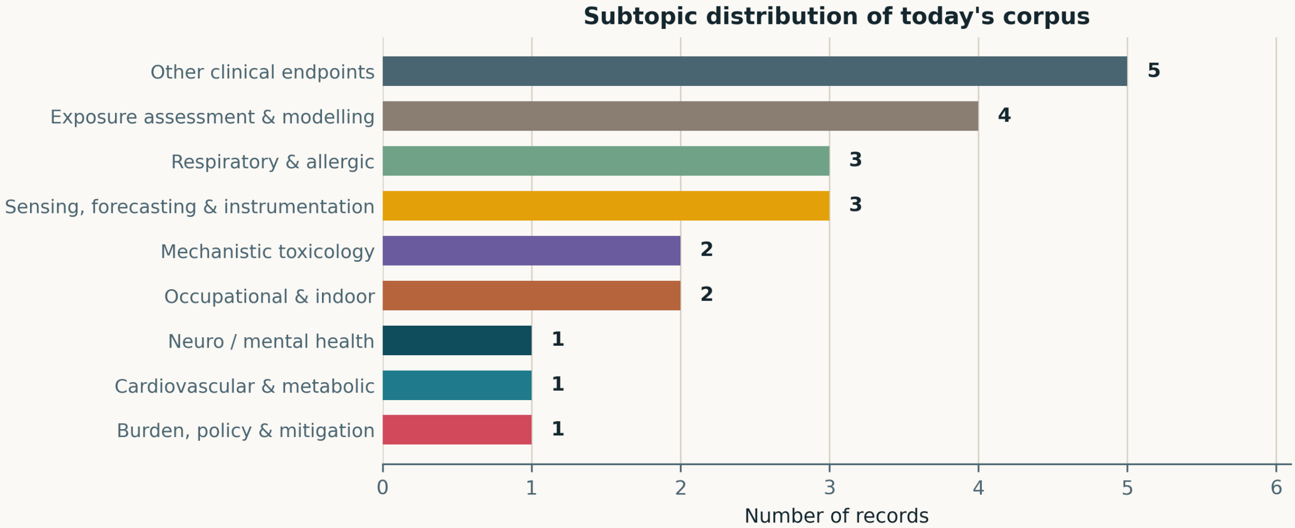

Fig. Subtopic assignment of the 22 in-scope records. Nine of the ten canonical clusters are represented — equalling 14 and 15 August for the widest spread since July, when three consecutive issues reached all ten. *Reproductive & developmental* is the single empty cluster, its ninth absence in twenty-five issues and the first since 13 August. Assignment is single-label by dominant contribution.

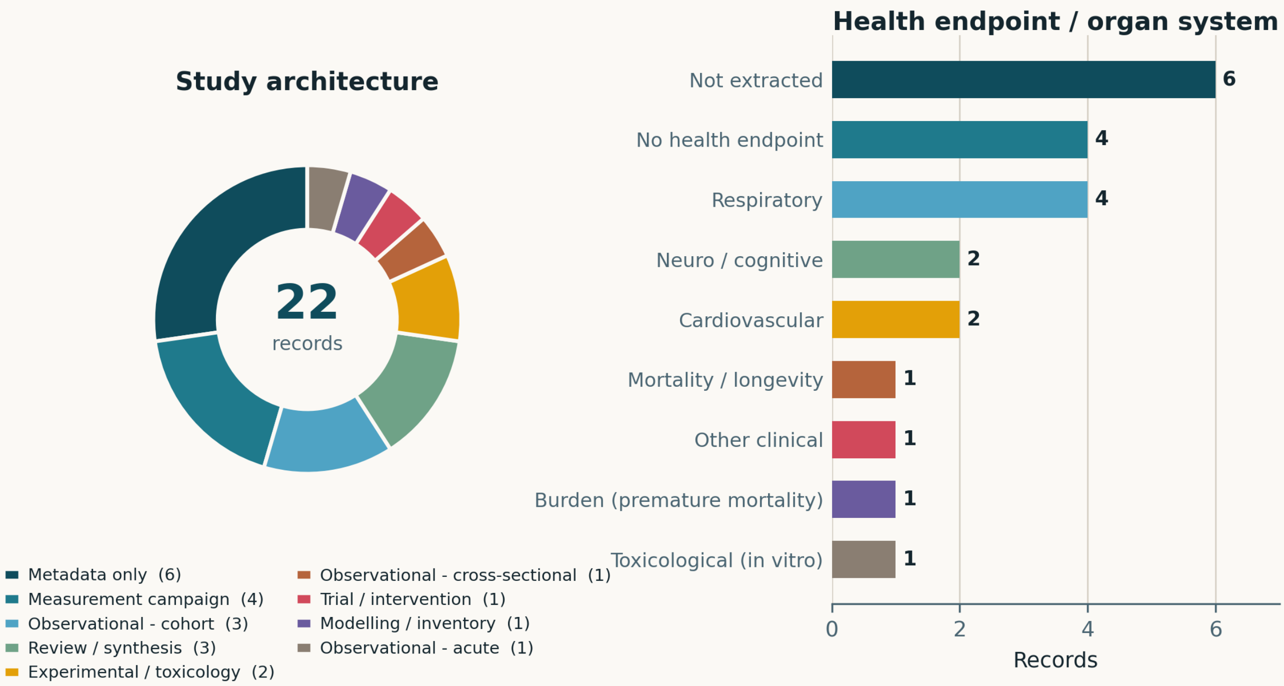

Fig. Left — study architecture. *Metadata only* is the largest single class at 6/22 (27%), the third-highest share of the watch after 8 August (43%) and 1 August (32%); these are publisher deposits that carried no abstract, not a study design. Measurement campaigns (4) outnumber observational cohorts (3). Right — health endpoint. Six records have no extractable endpoint because no abstract was deposited, and a further four genuinely carry none (one filtration trial, one emission-factor study, two aerosol-chemistry campaigns). Respiratory endpoints lead the extractable set at four.

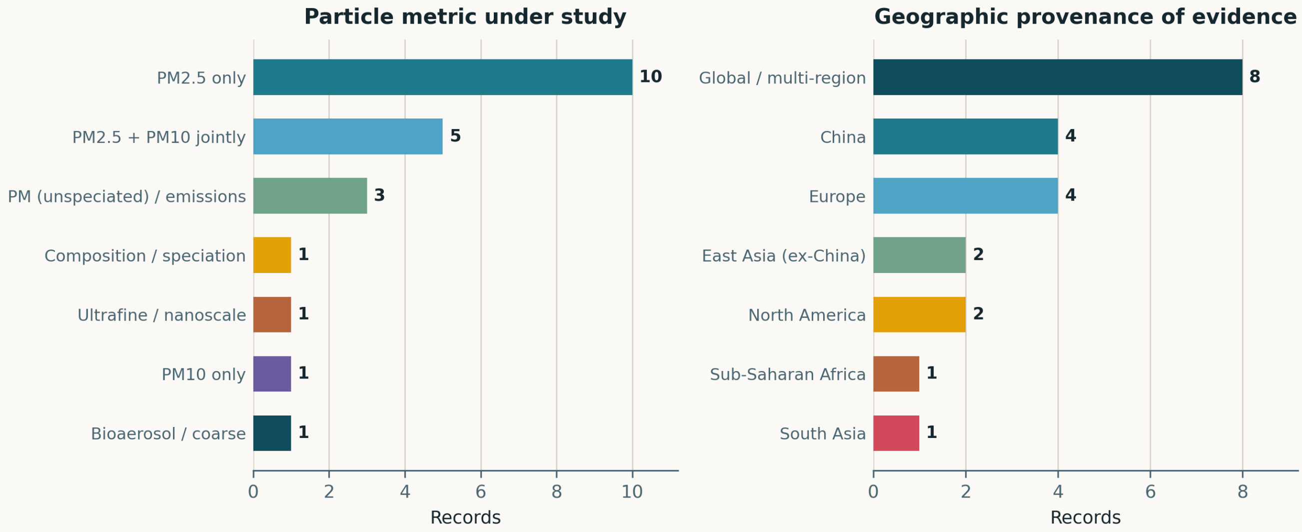

Fig. Left — particle metric. $PM_{2.5}$ alone is the exposure in 10 records and appears jointly with PM_{10} in a further 5. One record takes ultrafine particle *number* as its primary metric (a ten-year Athens size-distribution record) and one takes composition/speciation. Right — geographic provenance. The *Global / multi-region* bar aggregates three review syntheses, three records whose study location the abstract does not state, and two bench studies with no geography. Among located work, China and Europe tie at four; Sub-Saharan Africa (Addis Ababa) and South Asia (Lahore) contribute one each.

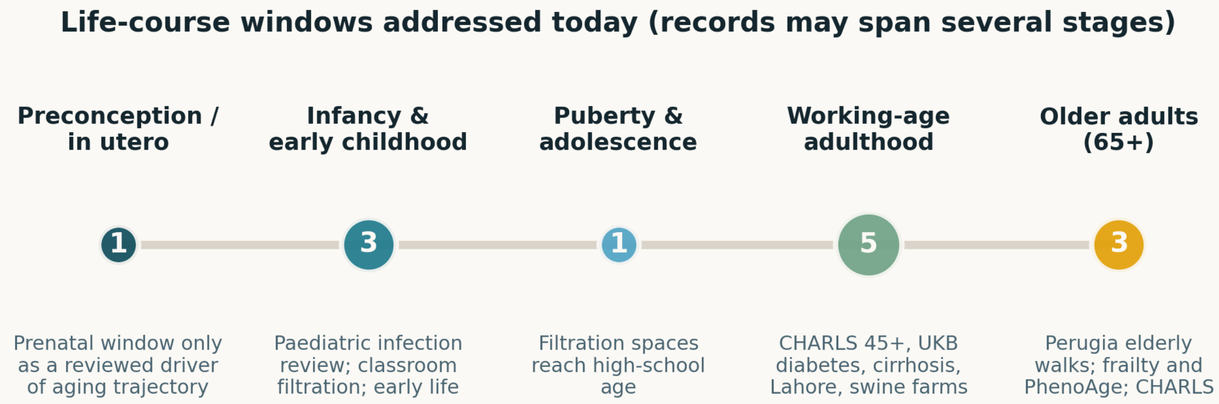

Fig. Life-course windows addressed. Counts are non-exclusive and a review counts once per window it explicitly treats. Working-age adulthood dominates (5) — the CHARLS 45+ wave, the UK Biobank diabetes cohort, the Korean cirrhosis registry, the Lahore outpatient sample and the swine-farm workers all sit in that band. The single in-utero entry is a reviewed determinant of biological-aging trajectory, not a primary prenatal-exposure study — today's corpus contains none.

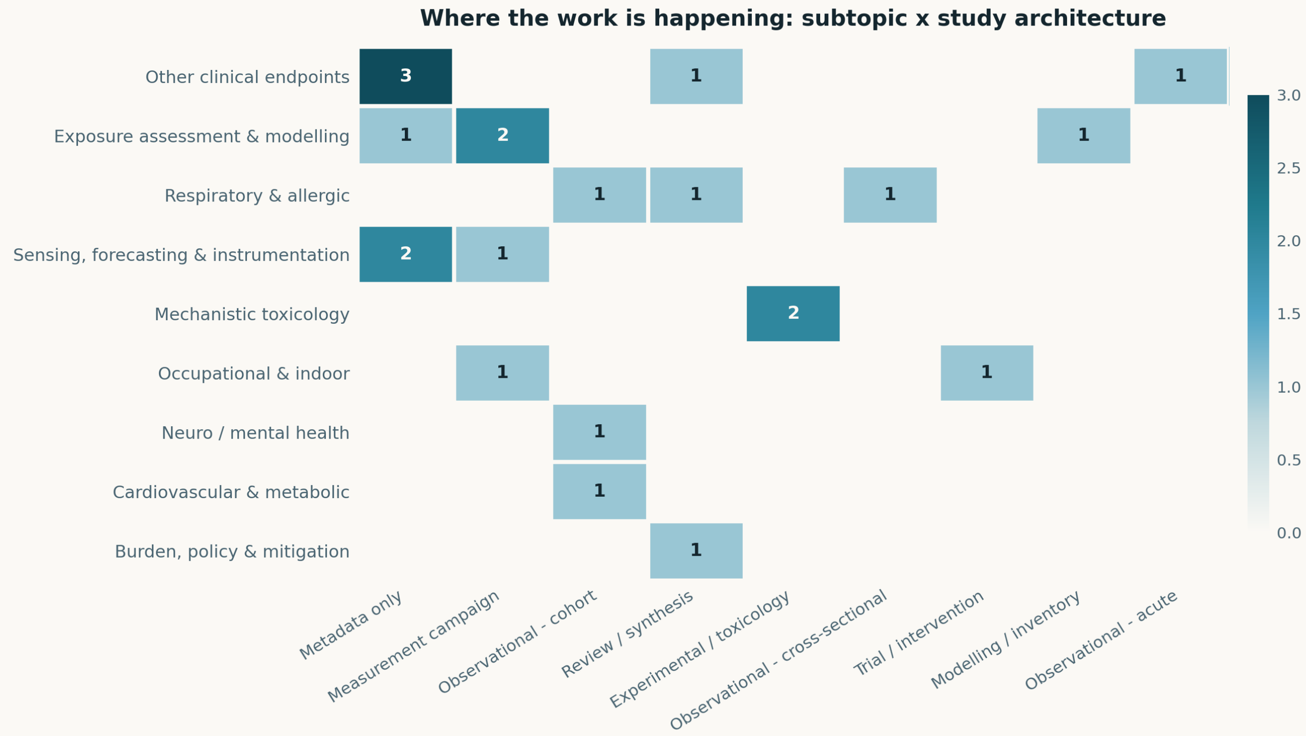

Fig. Subtopic × study-architecture cross-tabulation. Two structural features stand out. First, the *Metadata only* column is the most populated in the panel and spans three different subtopics — the abstract-free deposit problem is not concentrated in one literature. Second, the *Sensing, forecasting & instrumentation* row again contains no health-endpoint linkage of any kind, and *Exposure assessment & modelling* carries exactly one, the Perugia wearable study. That siloing is real today but is not structural: twelve of the nineteen August issues, this one included, have carried at least one sensing- or exposure-side record with a human health endpoint.

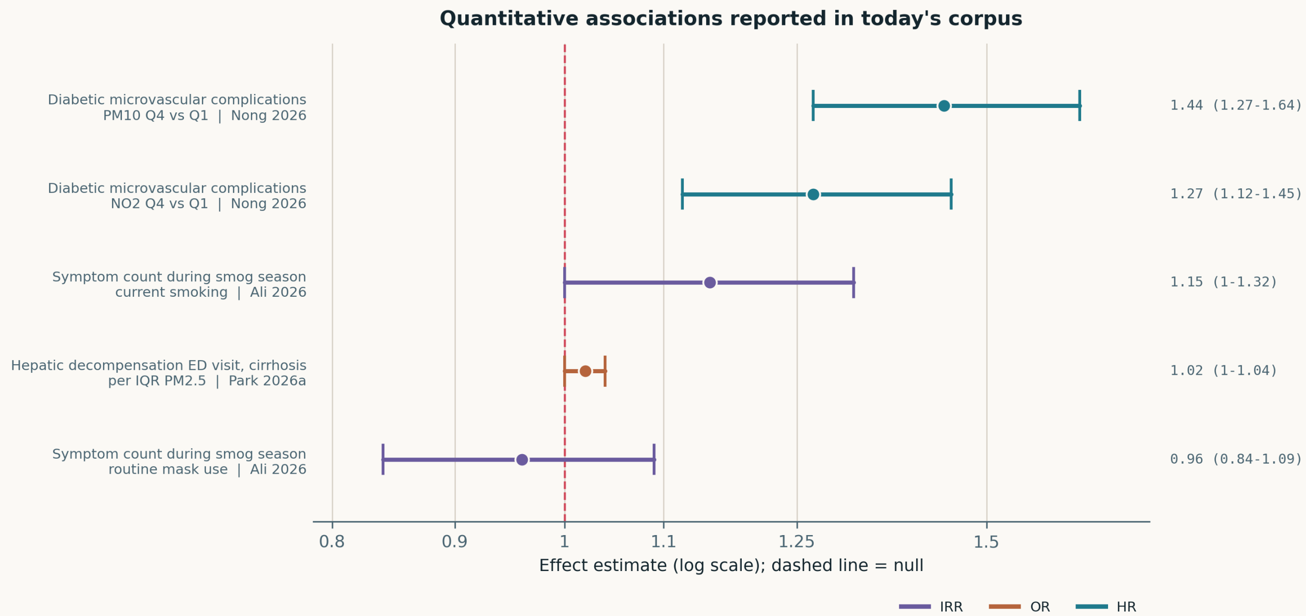

Fig. The five quantitative estimates reported in today's corpus, on a common logarithmic axis. **The exposure contrasts are not mutually comparable**: Nong et al. report top-versus-bottom quartile hazard ratios over a 12.4-year median follow-up, Park et al. a per-IQR odds ratio on a single day, and the two Ali et al. estimates are incidence rate ratios for *behavioural* covariates (smoking, mask use) measured during — not contrasts in — particulate exposure. Read direction and precision only. Note that the tightest interval belongs to the estimate closest to the null (Park, 29,167 patients) and that both Ali intervals cross or abut it on an n of 400.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Six records are marked METADATA ONLY: the publisher deposited no abstract, so nothing beyond title, journal and DOI is asserted for them.

Neuro, cognition & mental health — 1 record

He et al. 2026 Ecotoxicol Environ Saf

Plasma proteomic signatures modifying the air pollution–epilepsy association

Proteome-wide interaction screen: 51,428 participants, 2,254 Olink-style plasma proteins, $PM_{2.5}$/PM_{coarse}/NO_2/NO from land-use regression, plus a PCA-derived composite air-pollution index. Logistic models for prevalent and Cox models for incident epilepsy identify **CIT–$PM_{2.5}$, PTPRR–NO_2 and BCL2L15–API interactions**, enriched for immune, inflammatory and cell-death pathways, with hippocampal-volume and white-matter-hyperintensity correlates and neuronal/endothelial single-nucleus enrichment. The threat to inference is multiplicity: ∼2,254 proteins × four exposures with no corrected significance level or interaction magnitude reported in the abstract. A candidate list, not a mechanism. Relevance here is that it is exposure *misclassification* in the LUR layer, not protein measurement, that would most easily generate this pattern.

doi: 10.1016/j.ecoenv.2026.120669

Cardiovascular & metabolic — 1 record

Nong et al. 2026 PLoS One

Long-term air pollution and microvascular complications in diabetes

UK Biobank, 9,671 participants with diabetes, 12.4-year median follow-up, 2,104 (21.8%) incident microvascular complications. Cox models adjusted for demographic, clinical and biochemical covariates give **PM_{10} HR 1.44 (1.27–1.64)** and **NO_2 HR 1.27 (1.12–1.45)** for the highest versus lowest exposure quartile, stable across stratified and sensitivity analyses, with early divergence of the cumulative-incidence curves. Two caveats: $PM_{2.5}$ is listed among the exposures but no $PM_{2.5}$ estimate is reported in the abstract, and UK Biobank exposure contrasts are narrow, so the quartile framing hides how small the underlying concentration difference is. Restricted cubic splines are reported as showing risk beginning at relatively low levels — the claim that matters for guideline-adjacent argument, and the one the abstract quantifies least.

doi: 10.1371/journal.pone.0354711

Respiratory & allergic — 3 records

Ali et al. 2026 Public Health Chall

Symptom burden among adult outpatients during Lahore's 2024–2025 winter smog

400 outpatients at a tertiary hospital, October 2024 – January 2025, structured questionnaire plus city-level air quality (mean AQI 180, mean $PM_{2.5}$ 96 $\mu g/m^3$; $PM_{2.5}$ and PM_{10} above the WHO 24 h guideline on every monitored day). 98% reported at least one symptom — breathlessness 74%, eye irritation 56%, cough 53%. Regression on a composite vulnerability index: **smoking IRR 1.15 (1.00–1.32)**, **routine mask use IRR 0.96 (0.84–1.09), i.e. null**. Exposure is assigned at city level to a health-seeking sample with no unexposed comparison, so no dose–response can be read from this; its value is documenting symptom prevalence in a South Asian megacity where hospital-based evidence is thin.

doi: 10.1002/puh2.70349

Yang et al. 2026a J Asthma

Atmospheric humidity and air pollutants versus self-reported chronic lung disease

16,204 CHARLS adults aged 45+, 2011/2013/2015 waves, six pollutants and six humidity indices at 1 km. Each IQR increase in pollutant concentration carried **18–45% higher odds** of self-reported CLD; PM_{10} alone showed an approximately U-shaped exposure–response, and relative humidity an inverted-U, with a significant RH×sex interaction. Weighted quantile sum models were positive for both the pollutant and the humidity mixture. Self-reported outcome and a wide odds range quoted without per-pollutant intervals limit what can be taken from the point estimates; the reason to read it is the humidity treatment — six distinct indices rather than RH alone, which is the same distinction that separates a well-corrected from a badly-corrected low-cost nephelometer.

doi: 10.1080/02770903.2026.2721168

Rice & Miller 2026 Pediatr Pulmonol

Air pollution, microbes and respiratory infection risk in children

Narrative review joining two literatures usually kept apart: pollutant-driven epithelial barrier loss, oxidative stress and impaired innate/adaptive immunity, and the developing airway microbiome as a determinant of colonisation resistance. Argues that dysbiosis (antibiotics, air pollution, other perturbations) and pollutant exposure are partly substitutable routes to the same susceptibility phenotype, and notes that exposure to diverse environmental microbes — certain fungi included — may be protective. No quantitative synthesis and no bias assessment; useful as a framing citation for why an indoor sensing deployment in a childcare setting should log more than mass.

doi: 10.1002/ppul.71798

Mechanistic toxicology — 2 records

Sivakumar & Kurian 2026 Environ Toxicol

Duration-dependent transition from mitochondrial adaptation to persistent cardiac toxicity

Female Wistar rats, whole-body $PM_{2.5}$ at 250 $\mu g/m^3$, 3 h/day for 1, 7, 14 or 21 days, with cardiac mitochondrial bioenergetics, redox balance, quality-control signalling and ex vivo function assessed after each duration and after a 24 h washout. **1–7 days: adaptive and fully reversible. ≥14 days: suppressed respiration and ETC complex activity, mtDNA depletion, impaired biogenesis, intramitochondrial metal accumulation — none recovered within the washout.** Concentration was held constant, so duration alone moved the system across the threshold. The 24 h washout is short, so "persistent" means "not recovered in 24 h", which the authors state plainly; a longer recovery arm would settle it. The instrumentation implication is direct: a cumulative-duration-above-threshold metric is testable with continuous logging and invisible to a 24 h filter sample.

doi: 10.1002/tox.70189

Chen et al. 2026a J Chin Med Assoc

$PM_{2.5}$ suppresses IL-8-mediated epithelial migration in conjunctival cells

Immortalised human conjunctival epithelial line exposed to $PM_{2.5}$ with viability, apoptosis, scratch-migration and cytokine-array readouts. $PM_{2.5}$ reduced viability time- and dose-dependently with only mild apoptosis, but markedly impaired migration and wound closure. Cytokine profiling showed *selective* remodelling rather than a general inflammatory response, with IL-8 the most strongly suppressed mediator; recombinant IL-8 rescued migration and partially rescued viability, and IL-8 neutralisation impaired migration at baseline. Single immortalised line, suspension dosing, no dose anchored to an ambient concentration — so the mechanism is credible and the dosimetry is not. The ocular surface remains an under-instrumented exposure site.

doi: 10.1097/JCMA.0000000000001418

Occupational & indoor exposure 2 records

Xia et al. 2026 Front Pediatr

Enhanced filtration for indoor air quality in child-serving facilities

Week-long baseline and follow-up measurements in 33 spaces across 12 facilities spanning daycare to high school, with staff surveys, as part of a community-based participatory partnership. Air change rates were low throughout (0.9–1.4 h^{-1}); baseline school-hours indoor $PM_{2.5}$ 2.2–6.3 $\mu g/m^3$ and PM_{10} 5.7–26.9 $\mu g/m^3$, highest in standard schools. Enhanced filtration cut $PM_{2.5}$ **45–54%** in standard schools and the nursery and had little effect in childcare houses; a linear mixed model attributes ∼**51%** of the $PM_{2.5}$ reduction to the portable air cleaners. **Single-zone mass-balance prediction held for regular classrooms and the nursery but failed for portable buildings, very large rooms and repurposed houses.** That failure boundary is the most transferable result in the paper: it maps exactly onto where a deployed sensor earns its keep. No health endpoint was measured.

doi: 10.3389/fped.2026.1875263

Park et al. 2026b Environ Pollut

Phenotypic resistance, not community turnover, tracks antimicrobial use in swine-farm bioaerosol

962 isolates across six species, six Korean swine farms stratified by defined daily dose for animals, four compartments (faecal, aerosol, pig nasal, worker nasal) and two seasons, with flow-cytometric viability and qPCR-corrected abundance feeding a Monte Carlo QMRA. *S. aureus* isolate-level resistance was higher on high-usage farms; the other four species and the community composition showed no usage association. *Escherichia* abundance fell ∼four orders of magnitude on aerosolisation while *Staphylococcus* persisted and peaked in worker nasal samples. Median daily colonisation probability $1.98–8.63\times10^{-4}$; the model was most sensitive to aerosol *genus composition*, not total bioaerosol load. Six farms cannot resolve a farm-level effect and the authors say so. For sensing: total particle or total bioaerosol load is the wrong exposure variable here.

doi: 10.1016/j.envpol.2026.128991

Sensing, forecasting & instrumentation 3 records

Yang et al. 2026b Atmos Chem Phys

Dust transport versus local emissions in shaping urban ice-nucleating particles

Month-long winter campaign in Taiyuan, a heavily industrialised Chinese city: immersion-mode INP concentration and ice-nucleation active-site density alongside size distributions and chemical composition. N_{INP} 0.05–13.37 L^{-1} at and above $-15\,°\mathrm{C}$. During a desert-dust event **7.47 L^{-1} (95% CI 6.64–8.41) versus 0.61 L^{-1} otherwise**, n_s $1.57{\times}10^{7}$ versus $9.52{\times}10^{5}$ m^{-2}. During pollution periods N_{INP} correlated only weakly with sulfate, nitrate and OC ($|r| < 0.3$), and none of five PMF-resolved $PM_{2.5}$ source factors — industrial, dust, secondary, coal and traffic, fireworks — covaried significantly with it despite dominating the mass. One city, one winter, one dust event carrying the contrast. Still the cleanest recent demonstration that a $PM_{2.5}$ mass record does not index a co-located particle property.

doi: 10.5194/acp-26-11771-2026

Kalkavouras et al. 2026 Atmos Pollut Res

Ten-year trends in ultrafine particle number concentration and size distribution, Athens

METADATA ONLY — no abstract was deposited, so no trend, magnitude or instrument detail is asserted here. Flagged because a decade-long urban particle-number and size-distribution record is exactly the reference series that a low-cost optical network cannot reproduce and must be evaluated against; worth retrieving in full.

doi: 10.1016/j.apr.2026.103181

Ayele et al. 2026 Environ Sci Atmos

Fine-scale black carbon and TSP composition at a heavy-traffic intersection, Addis Ababa

METADATA ONLY — the deposited abstract is truncated to its opening sentences and reports no quantities. What is verifiable from it: the study addresses spatiotemporal variability of equivalent black carbon and the molecular composition of organics in total suspended particulate at a roadside intersection, and states that Ethiopian urban aerosol remains poorly characterised. Today's only sub-Saharan record.

doi: 10.1039/d6ea00100a

Exposure assessment & modelling 4 records

Blanco-Alegre et al. 2026 Atmos Chem Phys

Emission factors and particle morphology from prescribed fires in Mediterranean shrubland

Field campaign on *Calluna vulgaris* and *Genista hispanica* burns in north-west Spain combining gravimetry, thermal–optical transmittance, ion chromatography, FTIR and SEM. *Calluna* burned flaming (MCE 90.6%), *Genista* smouldering (MCE 70.8%); gas-phase concentrations were higher for *Calluna* but emission factors for incomplete-combustion products were higher for *Genista*. **$PM_{2.5}$ EFs 53.5 $\mathrm{g\,kg}^{-1}$ (*Calluna*) and 44.7 $\mathrm{g\,kg}^{-1}$ (*Genista*)**, with OC and EC at 28.1% and 32.9% of $PM_{2.5}$ mass and water-soluble inorganic ions at 6.9% and 4.5%. SEM showed abundant tar-ball-like submicron particles and thickly coated aggregates. Two vegetation types at one site — the EFs are not a regional average — but combustion-regime-resolved and directly usable.

doi: 10.5194/acp-26-11695-2026

Duan et al. 2026 Atmos Chem Phys

Regime-dependent aerosol pH in ammonia-rich urban Beijing from explainable machine learning

Field observations plus thermodynamic modelling, interrogated with interpretable ML to apportion pH variability. Temperature dominates, at roughly -0.6 **pH units per 10 °C**. Excess ammonia, nitrate-to-sulfate ratio and PM_1 mass are positively associated with pH, with strongest sensitivity at low values and saturation at high ones. Relative humidity is the interesting one: its sign *reverses* by regime — higher RH increases acidity below $\sim$15 °C, at N/S $>\sim$1.25, or at PM_1 $>\sim$50 $\mu\mathrm{g/m}^3$, and decreases it under warm, sulfate-dominant or low-mass conditions. Feature attribution is association, not causation, and pH is itself a model output rather than a measurement. The RH regime reversal is a useful caution for anyone applying a single humidity correction across seasons.

doi: 10.5194/acp-26-11683-2026

Mahia et al. 2026 J Environ Manage

Street geometry, air quality and real-time physiological response in elderly pedestrians

Fourteen elderly participants walked five routes in Perugia during spring while wearable sensors logged physiology and portable stations logged microclimate and air quality. Paths with higher height-to-width ratios (0.92–1.27) and elevated NO_2 and PM_{10} (to 21 ppb and 18.77 $\mu\mathrm{g/m}^3$) significantly degraded air-quality perception ($\beta = -0.273$, $p = 0.028$ for NO_2; $\beta = -0.312$, $p = 0.013$ for PM_{10}), and both pollutants tracked skin-conductance change and self-reported emotional state; higher Green Visual Index went with more relaxed responses. With $n = 14$ over one season the coefficients carry little weight, and perception and physiology are both plausibly driven by street geometry directly rather than by the pollutant. Read it for the measurement stack — paired personal air-quality and physiological logging at walking cadence — not the estimates.

doi: 10.1016/j.jenvman.2026.130681

Dovrou et al. 2026 Environ Sci Atmos

Wintertime wildfires reshape urban atmospheric chemistry: the January 2025 Los Angeles fires

METADATA ONLY — the deposited abstract is truncated before any result. Verifiable from it: the study concerns how dominant chemical regimes and meteorology govern wildfire air-quality impacts in urban coastal regions, using the January 2025 Los Angeles fires. Flagged for retrieval given how much of this watch's recent health signal has come from source-resolved wildfire $PM_{2.5}$.

doi: 10.1039/d6ea00096g

Burden, policy & mitigation 1 record

Bhatta et al. 2026 Sci Total Environ

Outdoor air purification technologies: efficiency, sustainability and health impact

PRISMA-ScR review, 266 records screened to 28 studies across five technology domains. Reported bests: smog towers at **97.8% $PM_{2.5}$ and 98.4% PM_{10} removal under optimal conditions**; solar-chimney hybrids within 2–4% of dedicated systems while generating electricity; TiO_2-modified cementitious composites potentially carbon-negative over their lifecycle; a roadside hedge cutting leaf-surface $PM_{2.5}$ magnetic loading 82%. The health-economics figure — 93,200 deaths averted in China at a 35 $\mu g/m^3$ indoor target, positive net monetary benefit in 93% of cities — is modelled from *indoor* air cleaners used as an analogue for outdoor systems, which the authors flag and which should be read as a bound on plausibility, not an estimate. "Optimal conditions" removal efficiencies are chamber or near-field numbers and say nothing about neighbourhood-scale concentration. The review's own priority list — standardised performance metrics, long-term field validation — is the honest summary.

doi: 10.1016/j.scitotenv.2026.182205

Other clinical endpoints 5 records

Park et al. 2026a Gut Liver

Short-term $PM_{2.5}$ and acute hepatic decompensation in cirrhosis

Nationwide time-stratified case-crossover on the Korean NHIS database, 2006–2021: 29,167 cirrhosis patients presenting to emergency departments with ascites, variceal bleeding, hepatic encephalopathy, jaundice or hepatorenal syndrome; control days matched on month and weekday; daily $PM_{2.5}$ assigned on a 1 km residential grid; conditional logistic regression adjusted for temperature, humidity, holidays and weekday. **aOR 1.02 (1.00–1.04) per IQR increase**, with the top exposure quintile 5% above the bottom and *attenuation* at longer averaging windows. No subgroup interaction reached significance. The design controls time-invariant confounding well, but the lower bound sits on the null and the effect is small enough that residual exposure misclassification from grid-assigned concentration could account for it. A large, clean, honestly reported near-null.

doi: 10.5009/gnl260039

Poursafa & Jylhävä 2026 Ageing Res Rev

Air pollution and biological aging: mechanisms, biomarkers and reversibility

Review organised by biomarker class rather than by outcome, which is what makes it useful. Evidence is graded as strongest for *system-level composites* — Klemera–Doubal biological age, PhenoAge, frailty indices — and weakest, at meta-analytic level, for *epigenetic clocks*, which the authors attribute to heterogeneity in exposure assessment, design and target-tissue cell composition rather than to absent biology. Causal mediation across independent cohorts supports biological aging as a pathway from exposure to cardiovascular, neuropsychiatric and renal disease. Pollutants of concern are $PM_{2.5}$, $PM_{0.1}$, traffic-related pollution and household combustion. Named gaps: household air pollution and low-income settings are under-represented, and longitudinal multi-omic designs are scarce. Air-quality and clean-cooking improvements are reported to partially reverse the acceleration — the strongest claim in the paper and the one most dependent on the intervention literature it summarises.

doi: 10.1016/j.arr.2026.103297

Wang et al. 2026 Atmos Environ

Long-term $PM_{2.5}$, aging hallmarks and age-related disease

METADATA ONLY — no abstract deposited. Title indicates a nationwide prospective cohort linking long-term $PM_{2.5}$ to aging hallmarks and age-related disease incidence. Cross-reads directly with Poursafa & Jylhävä above, which is the reason to retrieve it in full rather than wait for indexing.

doi: 10.1016/j.atmosenv.2026.122313

Lei et al. 2026 Atmos Environ

Hospital admission risk and burden of acute appendicitis attributable to $PM_{2.5}$

METADATA ONLY — no abstract deposited. Title indicates nationwide Chinese evidence on $PM_{2.5}$ and acute appendicitis admissions with an attributable-burden component. No effect size, lag structure or exposure model is asserted here.

doi: 10.1016/j.atmosenv.2026.122311

Chen et al. 2026b Atmos Environ

Air pollution and sarcopenia: meta-analysis and network toxicology

METADATA ONLY — no abstract deposited. Title indicates a meta-analysis paired with a network-toxicology arm. Pooled estimate, heterogeneity and included-study count are all unknown from the deposit; a meta-analysis carried without its numbers is a placeholder, recorded so it is not missed on a later sweep.

doi: 10.1016/j.atmosenv.2026.122312

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
He et al.	Large-scale plasma proteomics identifies protein signatures linking air pollution to epilepsy	*Ecotoxicol Environ Saf*	Prospective cohort	$PM_{2.5}$, PM_{coarse}, NO2, NO from land-use regression	Not stated
Cardiovascular & metabolic					
Nong et al.	Air pollution and diabetic microvascular complications: a prospective cohort study	*PLoS One*	Prospective cohort	PM_{10}, $PM_{2.5}$, NO2 (annual, quartiles)	UK
Respiratory & allergic					
Ali et al.	Symptom burden among adult outpatients during the 2024-2025 winter smog season in Lahore	*Public Health Chall*	Cross-sectional (multistage)	City-level $PM_{2.5}$ (mean 96 $\mu g/m^3$), PM_{10}, CO, AQI	Pakistan
Rice & Miller	Air pollution, microbes and respiratory infection risk in children	*Pediatr Pulmonol*	Narrative review	$PM_{2.5}$ and indoor combustion PM (qualitative)	Global
Yang et al.	Interactive associations of atmospheric humidity and air pollutants with self-reported chronic lung disease	*J Asthma*	Prospective cohort (mixtures)	$PM_{2.5}$, PM_{10} + 4 gases at 1 km, with 6 humidity indices	China
Mechanistic toxicology					
Chen et al.	$PM_{2.5}$ suppresses IL-8-mediated epithelial migration of human conjunctival epithelial cells	*J Chin Med Assoc*	In vitro (cell lines)	$PM_{2.5}$ (suspension, time- and dose-response)	Laboratory
Sivakumar & Kurian	Repeated $PM_{2.5}$ inhalation drives a duration-dependent transition from mitochondrial adaptation to persistent cardiac toxicity	*Environ Toxicol*	Animal model	$PM_{2.5}$ 250 $\mu g/m^3$, 3 h/-day, 1-21 days (whole-body inhalation)	Laboratory
Occupational & indoor					
Park et al.	Within-species phenotypic resistance rather than community turnover tracks antimicrobial usage in swine-farm bioaerosol	*Environ Pollut*	Environmental surveillance campaign	Viable bioaerosol abundance (flow cytometry + qPCR-corrected)	South Korea
Xia et al.	Effectiveness of enhanced filtration interventions for indoor air quality in child-serving facilities	*Front Pediatr*	Trial / intervention	Indoor $PM_{2.5}$ and PM_{10}, week-long pre/post continuous	USA
Sensing, forecasting & instrumentation					
Ayele et al.	Fine-scale variations in black carbon and TSP chemical composition at a heavy-traffic intersection, Addis Ababa	*Environ Sci Atmos*	Metadata only (no abstract)	Equivalent black carbon + total suspended particulate organics	Ethiopia
Kalkavouras et al.	Long-term (10-year) trends in ultrafine particle number concentrations and size distributions in Athens	*Atmos Pollut Res*	Metadata only (no abstract)	Ultrafine particle number concentration + size distribution	Europe
Yang et al.	Dust transport and local anthropogenic emissions differently shape ice-nucleating particles in an industrial urban atmosphere	*Atmos Chem Phys*	Measurement campaign	Immersion-mode INP concentration, size distribution, $PM_{2.5}$ PMF factors	China
Exposure assessment & modelling					
Blanco-Alegre et al.	Emissions from prescribed fires in two Mediterranean shrublands: chemical and morphological analysis	*Atmos Chem Phys*	Field measurement	Gravimetric $PM_{2.5}$ emission factors + OC/EC, ions, SEM morphology	Spain
Dovrou et al.	Wintertime wildfires reshape urban atmospheric chemistry: evidence from the January 2025 Los Angeles fires	*Environ Sci Atmos*	Metadata only (no abstract)	Wildfire-influenced urban aerosol (not extractable)	USA
Duan et al.	Regime-dependent aerosol pH variability in ammonia-rich urban Beijing from explainable machine learning	*Atmos Chem Phys*	Machine-learning model	PM1 mass and composition + thermodynamic pH	China
Mahia et al.	Real-time effects of street geometry on elderly well-being under combined air-heat exposure	*J Environ Manage*	Observational exposure study	Personal portable PM_{10} (to 18.77 $\mu g/m^3$) and NO2, wearable physiology	Italy

First author	Focus	Journal	Design	Metric	Region
Burden, policy & mitigation					
Bhatta et al.	Outdoor air purification technologies for urban PM and gaseous pollutant mitigation: a comprehensive review	*Sci Total Environ*	Systematic review (PRISMA)	$PM_{2.5}$ and PM_{10} removal efficiency across five technology domains	Global
Other clinical endpoints					
Chen et al.	Air pollution and sarcopenia: a meta-analysis and network toxicology analysis	*Atmos Environ*	Metadata only (no abstract)	Air pollution mixture (not extractable)	Not stated
Lei et al.	Hospital admission risk and burden of acute appendicitis due to fine particulate matter pollution	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ (not extractable)	China
Park et al.	Short-term fine particulate matter and hepatic decompensation in cirrhosis: a nationwide case-crossover study	*Gut Liver*	Case-crossover	Daily $PM_{2.5}$ on a 1 km residential grid	South Korea
Poursafa & Jylhava	Aging in the air we breathe: mechanisms and consequences of air pollution exposure for biological aging	*Ageing Res Rev*	Mechanistic review	$PM_{2.5}$, PM0.1, TRAP, household combustion (qualitative)	Global
Wang et al.	Long-term exposure to $PM_{2.5}$, aging hallmarks and age-related diseases: a nationwide prospective cohort study	*Atmos Environ*	Metadata only (no abstract)	Long-term $PM_{2.5}$ (not extractable)	Not stated

Provenance and method

Entry window **19 August 2026**, contiguous with the 18 August issue. Sources queried on 19 August: **PubMed** E-utilities and the PubMed connector, run as two separate queries against `[Date - Entry]` — a health axis and a monitoring/sensing axis (18 unique PMIDs between them, 14 local and 4 connector-only); **Crossref by ISSN** across the instrumentation and atmospheric journals PubMed does not index (17 deposits over 18 journals); **Europe PMC** `CREATION_DATE` (0 records — a topic-free control query returned only 0 records of *any* kind created that day against 4 on 18 August and 2,794 on 17 August, so the index had not ingested the day rather than the leg being broken); **arXiv** relevance query with a three-day recency screen (0 in window, newest posting predates it); **Consensus** (no in-window record, tenth consecutive zero-carry); **OpenAlex** HTTP 429, seventeenth consecutive failure — that leg remains dead. ClinicalTrials.gov returned no new PM-related registration; `state/trials.json` is unchanged and is surfaced in the weekly rollup. Deduplication keys on PMID, then lowercased DOI, then a title/author/year signature, checked against the running `seen.json` index before any record is summarised; no duplicate was found today. **13 records were rejected** with a logged reason — greenhouse-gas and carbon-accounting work, cloud and gas-phase atmospheric chemistry, an e-cigarette aerosol study, and a commentary with no abstract — all in `state/rejected.jsonl`. **Six of the 22 in-scope records carry no abstract** because the publisher deposited none; they are marked METADATA ONLY in the digest and as "Metadata only" in the register and figures, and nothing beyond title, journal and DOI is asserted for them. Subtopic, design, particle-metric and geography labels are assigned from title, abstract and metadata; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Every DOI was machine-checked against its PubMed record and resolved before the build (0 fail, 0 warn). Content derived from publisher and PubMed metadata; abstracts remain the copyright of their respective publishers.